

Alternative Credit Scoring for Philippine Rural Banks

A proof-of-concept model built on World Bank Global Findex 2025 data

Project at a glance

Item	Result
Data source	World Bank Global Findex 2025 (144,090 survey cases)
Modelling dataset	9,691 loan applicants, 74:26 approved-to-declined
Retained features	78 predictors after screening from 199
Models compared	L1 (Lasso) Logistic Regression, CatBoost, XGBoost
Selected model	XGBoost, class-weighted, tuned for decline recall
Decline-class recall / F1	0.726 / 0.530 on a 1,939-record holdout
Inclusion variant recall	0.608 using non-traditional features only

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September 2026

I. Problem Statement

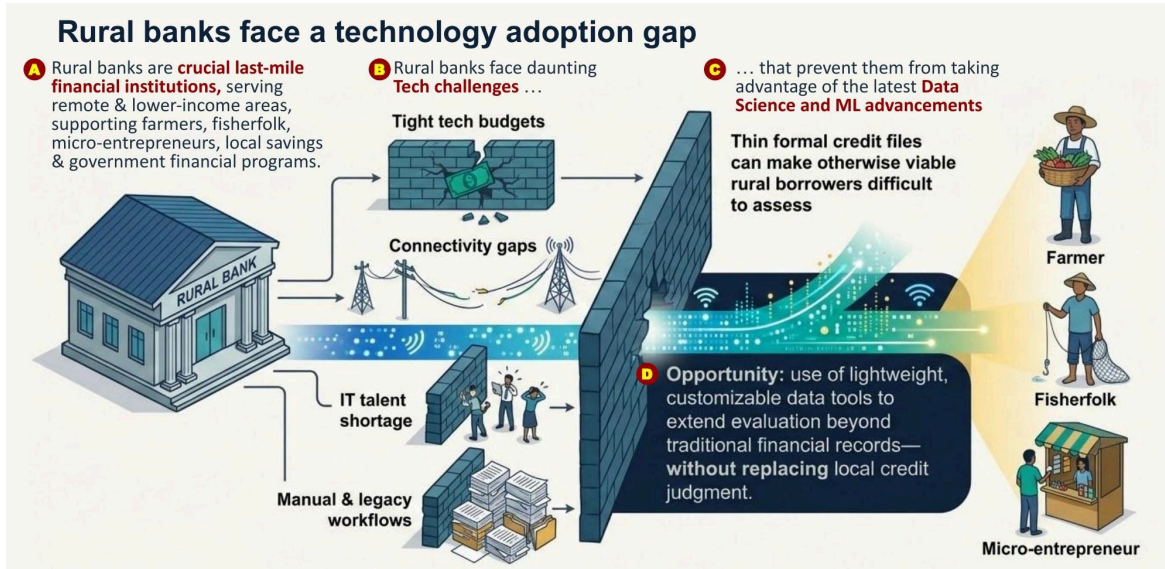


Figure 1. Rural banks face a technology adoption gap

1.1 Rural banks as last-mile lenders and their technology adoption challenges

The Rural Bankers' Association of the Philippines (RBAP) describes rural banks as last-mile channels for grassroots growth and financial inclusion. They operate in remote and lower-income municipalities that commercial banks treat as unviable. They have four vital functions:

- They **advance financial inclusion** by bringing basic services to unbanked populations.
- They **support countryside development** through credit to small-scale farmers, fisherfolk SMEs.
- They **serve as government conduits** for public financial aid and agricultural credit programs.
- They **mobilize local savings**, taking deposits and reinvesting them in the same community.

Rural banks trail their universal, commercial, thrift and digital counterparts in **technology adoption** because of these constraints: limited tech budgets, poor internet connectivity in remote areas, acute shortages of IT talent, & heavy reliance on legacy and manual systems.

That technology lag has a price. Larger lenders now score unbanked applicants using new metrics (financial platform use, utility payment logs & mobile top-ups); smaller rural banks still use traditional measures. An applicant with a thin formal credit file stays hard to assess, even if that farmer or store owner is creditworthy.

The project aims to build a proof-of-concept alternative credit scoring model for rural banks. The model draws on non-traditional parameters such as electronic money use, utility payments & mobile phone use. It serves as a lightweight framework that a bank can customize quickly to its own dataset, processes & risk appetite.

Credit scoring is a binary classification problem: approve or decline. The minority class carries the risk. A lender protects its portfolio by catching the applicants it should decline, so the project assessed models on minority-class metrics rather than overall accuracy (Table 1).

Table 1. Success metrics for the decline (minority) class

Metric	What it measures
Precision	Share of predicted declines that were actual declines
Recall	Share of actual declines that the model caught
F1 score	Harmonic mean of precision and recall, balancing both in one number

1.2 Business value

Alternative scoring creates value on both sides of the credit decision. On the approval side, it can surface creditworthy applicants who fail conventional thresholds and grow the loan book. On the decline side, it can flag risks that sparse traditional data hides and protect portfolio quality.

Approval uplift. Assuming an average loan portfolio of around ₱737 million per institution, a 10% 10% uplift in approvals, would produce indicative portfolio growth of about ₱74 million for an average rural bank.

Loss reduction. Historical overdue or default rates for rural banks run between 10% and 15%. Assuming a 2 percentage point reduction in past-due loans and NPLs applied to the average ₱737 million portfolio, that illustrative reduction is worth about ₱15 million.

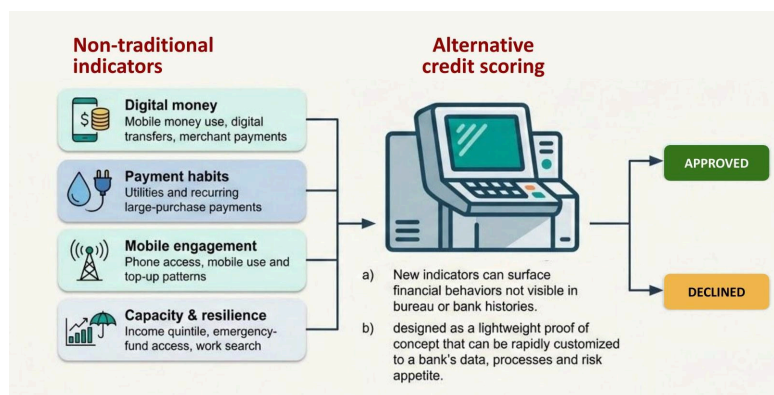


Figure 2. Machine learning can help close the information gap

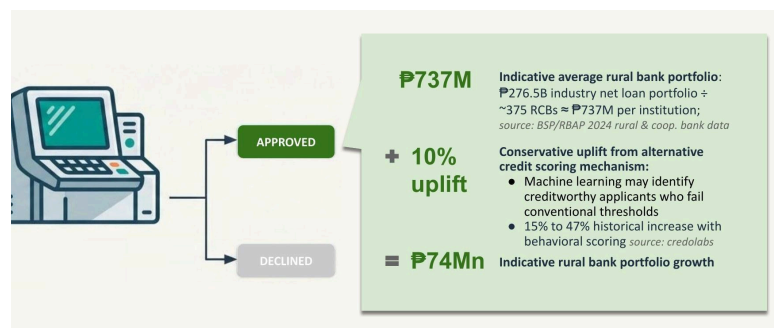


Figure 3. Approval uplift can translate into portfolio growth

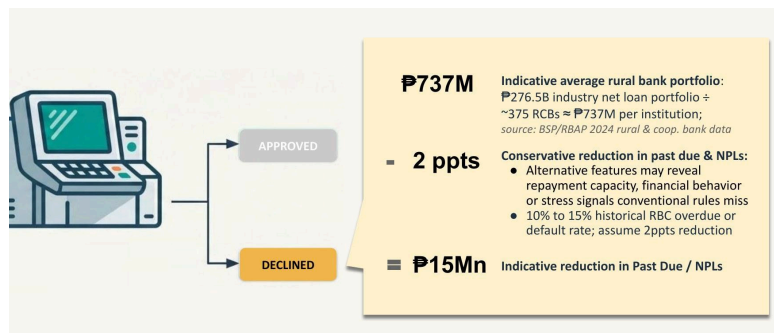


Figure 4. Better risk signals may reduce credit losses

II. Dataset

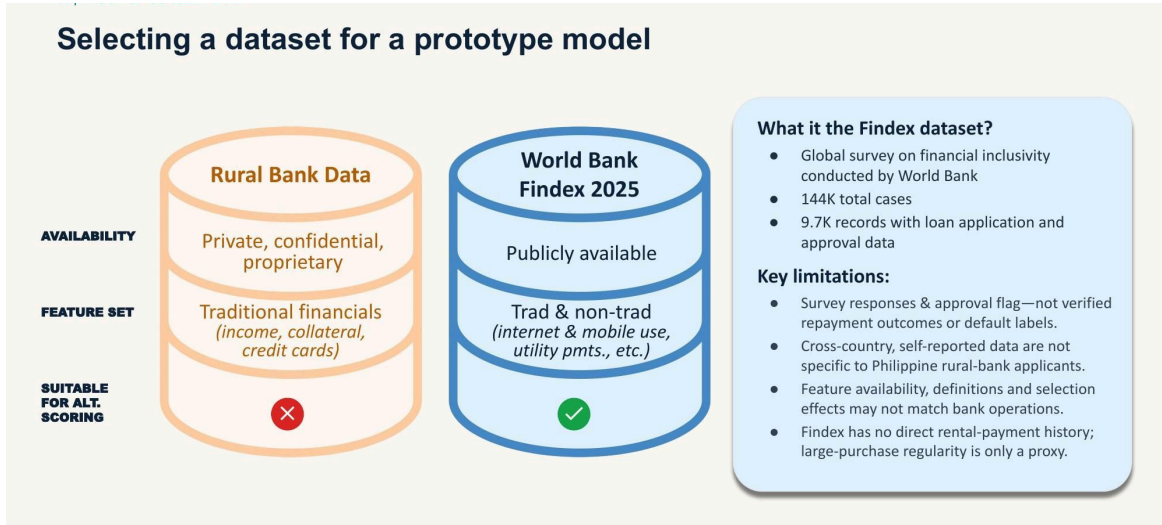


Figure 5. Selecting a dataset for a prototype model

2.1 Source

Banks keep loan application, approval and denial records private. To stand in for that data, the project used the World Bank's **Global Findex Database 2025: Connectivity and Financial Inclusion in the Digital Economy**, produced by the Development Research Group, Finance and Private Sector Development Unit. The microdata and data dictionary from the World Bank microdata catalog (study 7860) was downloaded on 22 Sept. 2026.

2.2 From a Philippine subset to the global superset

The Philippine subset of this data has only 64 records with loan application & approval data – too few to train a classifier. The global superset has 144,090 cases with 9,691 carrying a loan application flag with an approval or denial outcome, enough for model training. The trade-off is that a model trained on global survey is not directly usable by Philippine rural banks; fine-tuning or retraining with own data is needed prior deployment.

2.3 Variables & known limitations

Findex is a broad survey of financial behavior with 199 variables. Demographic variables cover country, population, age and similar fields. Traditional socioeconomic indicators include household income quintile, credit card ownership and savings. Alternative indicators, the largest group, include utility payments, frequency of digital transactions, and mobile phone and plan ownership.

Table 2. Findex 2025 variables by category

Category	Features
Demographic / psychographic	27
Socioeconomic, traditional	68
Socioeconomic, alternative	102
Others (survey parameters)	2
Total	199

Findex has a loan application and an approval flag. It does not record verified repayment outcomes or default labels. The data is cross-country & self-reported, not specific to Philippine rural-bank applicants. Feature definitions and selection effects may not match how a bank operates. Findex also lacks direct rental-payment history; the closest proxy is regular payment for a car, house or other large purchase (fin25e3).

III. EDA & Feature Engineering

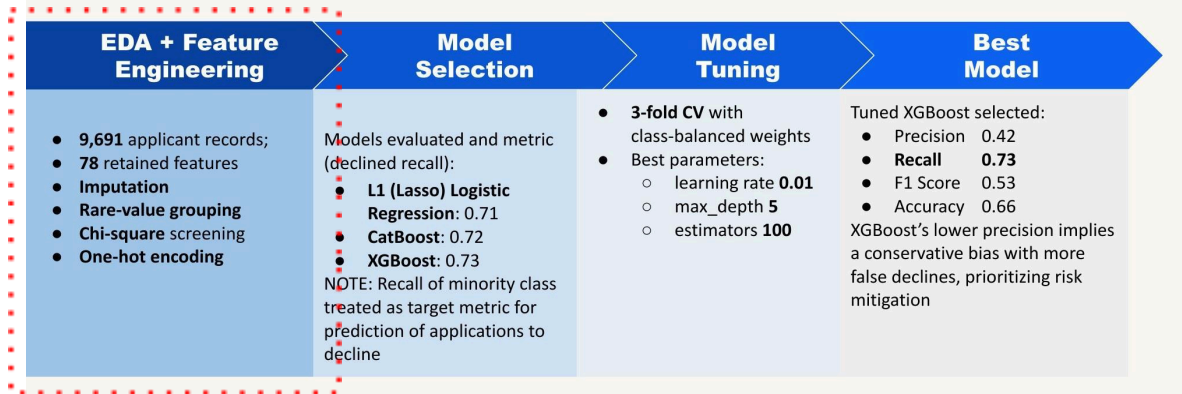


Figure 6. EDA, Feature Selection & Engineering in the Model build process

3.1 Row filtering and target

Only loan applicants (fin20) were considered, removing 93% of the records and leaving **9,691 rows**. The target variable, fin21, has 74 (approved):26 (declined) split.

3.3 Feature selection and engineering

Step 0: feature budget. Events Per Variable (EPV) rule of thumb to cap model size. With 26.2% minority class, and an EPV of 15 to 20, we have a ceiling of 132 to 169 features.

Step 1: prohibited and irrelevant attributes. Following the US Equal Credit Opportunity Act, attributes that could perpetuate bias (sex, age, country & region) were dropped. Survey constants such as year & fin20 (target) were also removed, along with irrelevant items (e.g. calamity occurrences). This dropped 21 columns.

Step 2: sparsity and imputation. A total of **91 columns with 50% or more NULL** values were dropped to reduce noise. Remaining features with NULL were imputed with supermajority value (if one value exceeded two-thirds of responses), or with a “Don’t know” code to avoid artificial signals.

Step 3: rare-value grouping. To control the number of one-hot encoding dummy columns, rare values (< 5% of responses) were folded into related categories, mostly “Don’t know” or “Did not”. **71 columns** re-binned.

Step 4: chi-square screening. A chi-square test of association measured each remaining variable against fin21, with every contingency table meeting the minimum expected frequency of 5. A liberal p-value threshold of 0.25 was used so as not to discard weak but potentially useful signals. **6 features** failed the test & were dropped.

Table 3. Feature funnel

Stage	Count
Beginning feature set	199
Less: prohibited, constant, irrelevant, plus ID, weight and target	(24)
Less: very sparse (50% or more null)	(91)
Less: statistically irrelevant (failed χ^2 test)	(6)
Final predictor set	78

The final 78 predictors sit well below the EPV ceiling.

IV. Modelling

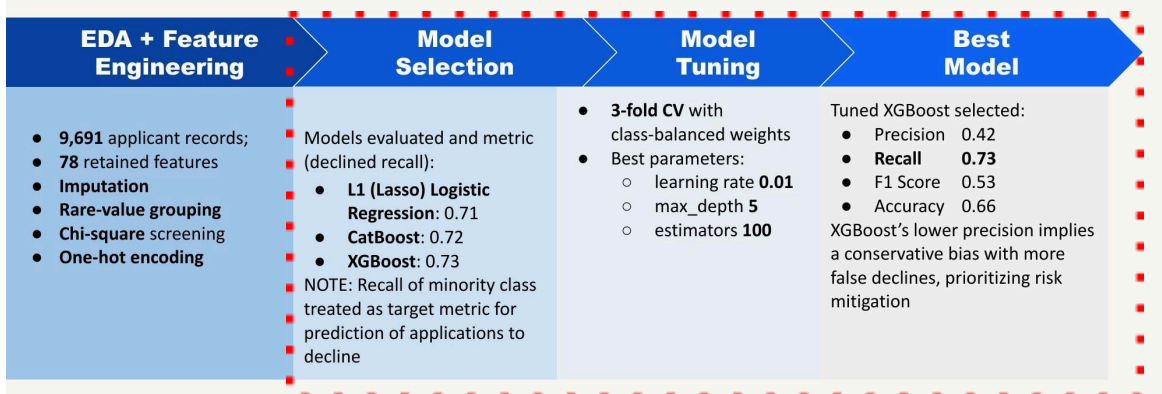


Figure 7. Model build & tuning post EDA & Feature selection

4.1 Setup & Baseline Results

A stratified 80/20 split produced **7,752 training records** and a **1,939-record test set**. The Index survey weights were retained through the split. Three algorithms were evaluated:

- **L1 (Lasso) Logistic Regression**, run inside a pipeline with one-hot encoding
- **CatBoost**, using categorical codes directly
- **XGBoost**, using categorical codes directly

Baseline models were trained per type of algorithm, all of which failed at the target metric of minority class recall: L1 (Lasso) LR captured only (24% decline recall), CatBoost (22%), XGBoost (31%)

4.2 Tuning

Each model received class weighting to offset the 74:26 imbalance. A grid search with 3-fold cross-validation then picked the settings with the highest **decline-class recall**, using a custom scorer.

- L1LR used balanced class weights & tested C across 0.01, 0.1, 1, 10 & 100; **C = 0.01** was optimal.
- CatBoost used auto balanced weights & settled on depth of 4, 100 iterations & a 0.05 learning rate.
- XGBoost used a scale_pos_weight of 2.80 & settled on a **0.01 learning rate, max depth of 5 and 100 estimators**.

In all three cases, the recall-driven search favored simpler, more heavily regularized settings.

4.3 Results & Model Selection

Table 5. Tuned models, decline-class metrics on the holdout

Model	Precision	Recall	F1	Accuracy
L1 Logistic Regression	0.424	0.712	0.531	0.669
CatBoost	0.431	0.720	0.539	0.676
XGBoost	0.417	0.726	0.530	0.661

Recall-driven tuning improved all three models. For example, decline recall for XGBoost rose from 0.31 to **0.726**, and its F1 from 0.38 to **0.530**. Its precision, however, declined from 0.48 to 0.42. Note that the spread between the three models is narrow: 1.4 points in recall and under 1 point in F1.

The **tuned XGBoost** model is the recommended option because it has the highest decline recall, our primary metric. It also has the lowest precision, which means more false declines and makes it the most conservative of the three. That bias suits small lenders with limited resources.

V. Model Audit

Audit shows most impactful characteristics — and an inclusion trade-off

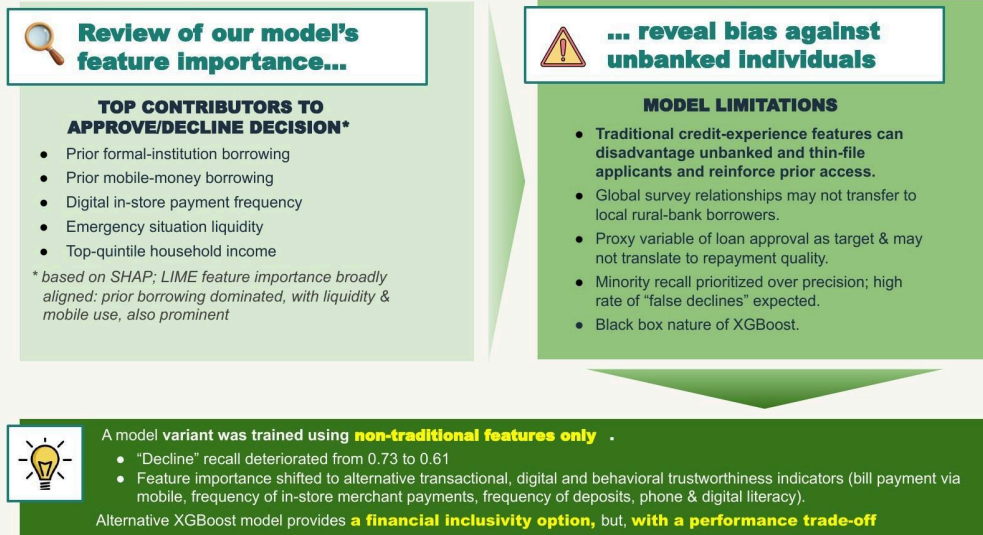


Figure 8. Model audit and implications

5.1 Feature importance

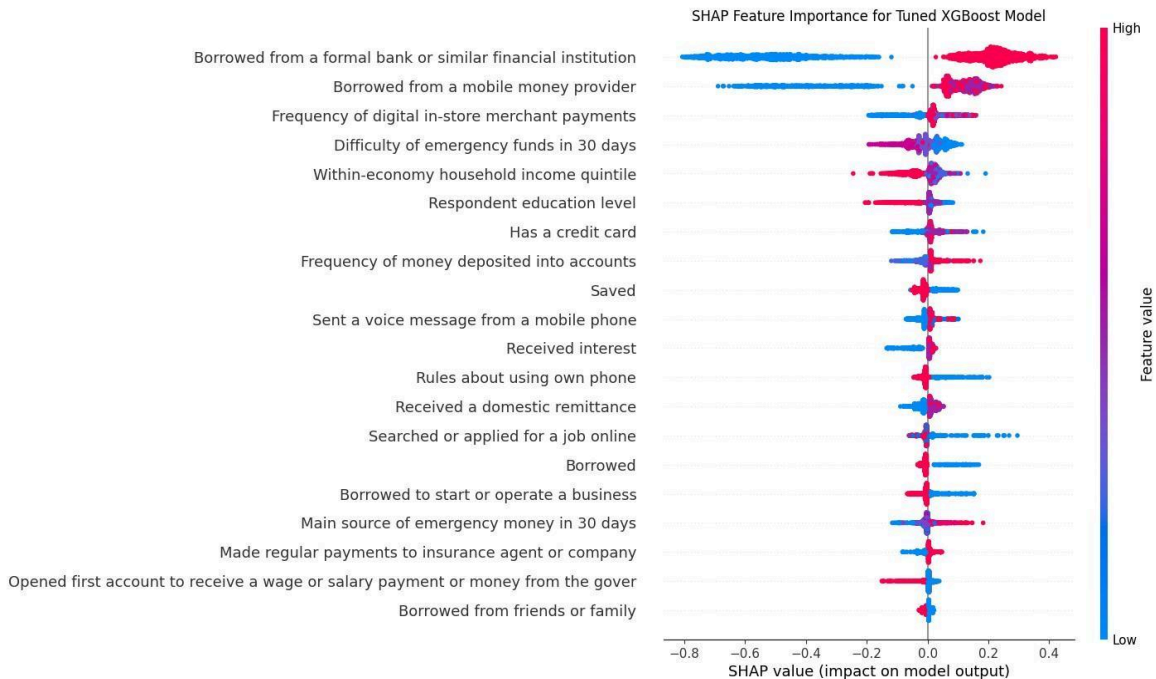


Figure 9. SHAP feature importance for the tuned XGBoost model (notebook \$5.i)

SHAP analysis (Figure 9) of the tuned XGBoost model yield the top predictors in our model:

1. prior borrowing from a formal financial institution
2. prior borrowing from a mobile money provider
3. frequency of digital in-store merchant payments
4. difficulty raising emergency funds w/in 30 days
5. household income quintile
6. respondent education level

The two borrowing features dominate. Applicants with credit history have higher chances of approval; those without prior borrowing records are predisposed to be declined. Higher income quintiles & higher education also favor approval. Prior experience with loans, liquidity & socioeconomic capacity drive the model's decisions.

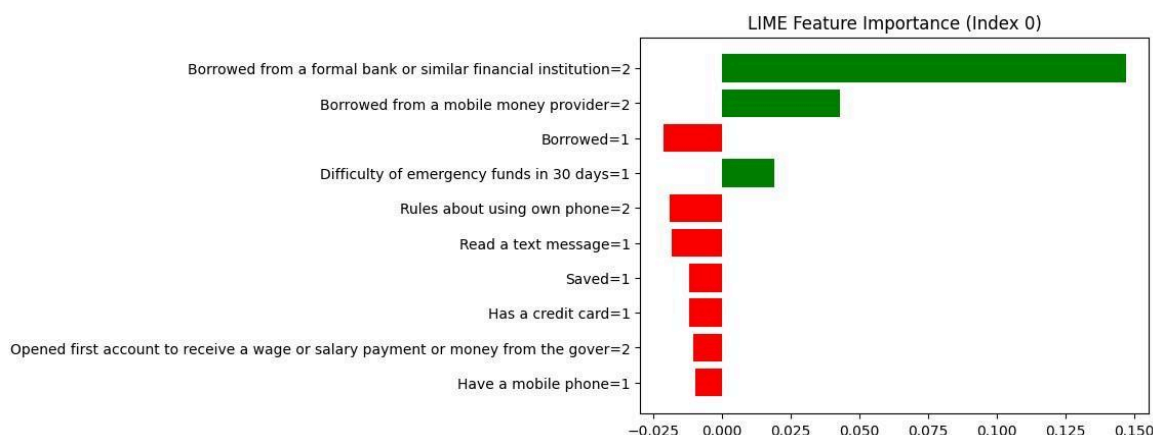


Figure 10. LIME explanation for one holdout applicant

LIME was conducted on a single holdout applicant as a cross-check (Figure 10). For that applicant, absence of credit history had the highest contribution to a decline decision, with emergency-fund liquidity and mobile phone use also contributing. LIME broadly corroborates the SHAP ranking.

5.2 Other Model limitations

1. Findex is a **global dataset**. It doesn't reflect the Philippine context nor rural-bank lending; the model must be fine-tuned or retrained on a bank's own data prior deployment.
2. We used a **proxy target (loan approval)**: the model learns to mimic historical approval decisions, rather than predict actual default risk.
3. We have a **recall-precision trade-off** that is deliberate but costly. Tuning to increase decline recall to 72.5% dropped decline precision to 41.7%. The model is prone to flagging many creditworthy applicants as high risk. This could restrict portfolio growth & deny credit to good borrowers.
4. Finally, XGBoost is a **black box**. Its set of non-linear splits means each credit decision needs a SHAP or LIME explanation to produce reason codes vs logistic regression that can be read from its coefficients.

5.3 Bias and fairness

Though the usual sources of bias (gender, age and ethnicity) were dropped upfront, the audit still revealed a major financial inclusivity problem: the features that emerged as most important are **prior credit experience** and **capacity to pay**. An unbanked applicant has no borrowing history to show, so the model reinforces existing access instead of advancing inclusion.

5.3.1 Alternative model

As an alternative, a model was trained on purely non-traditional attributes (13 traditional financial-experience features were removed, including: formal account ownership, borrowing and saving, deposits, etc.) A total of 65 attributes were left.

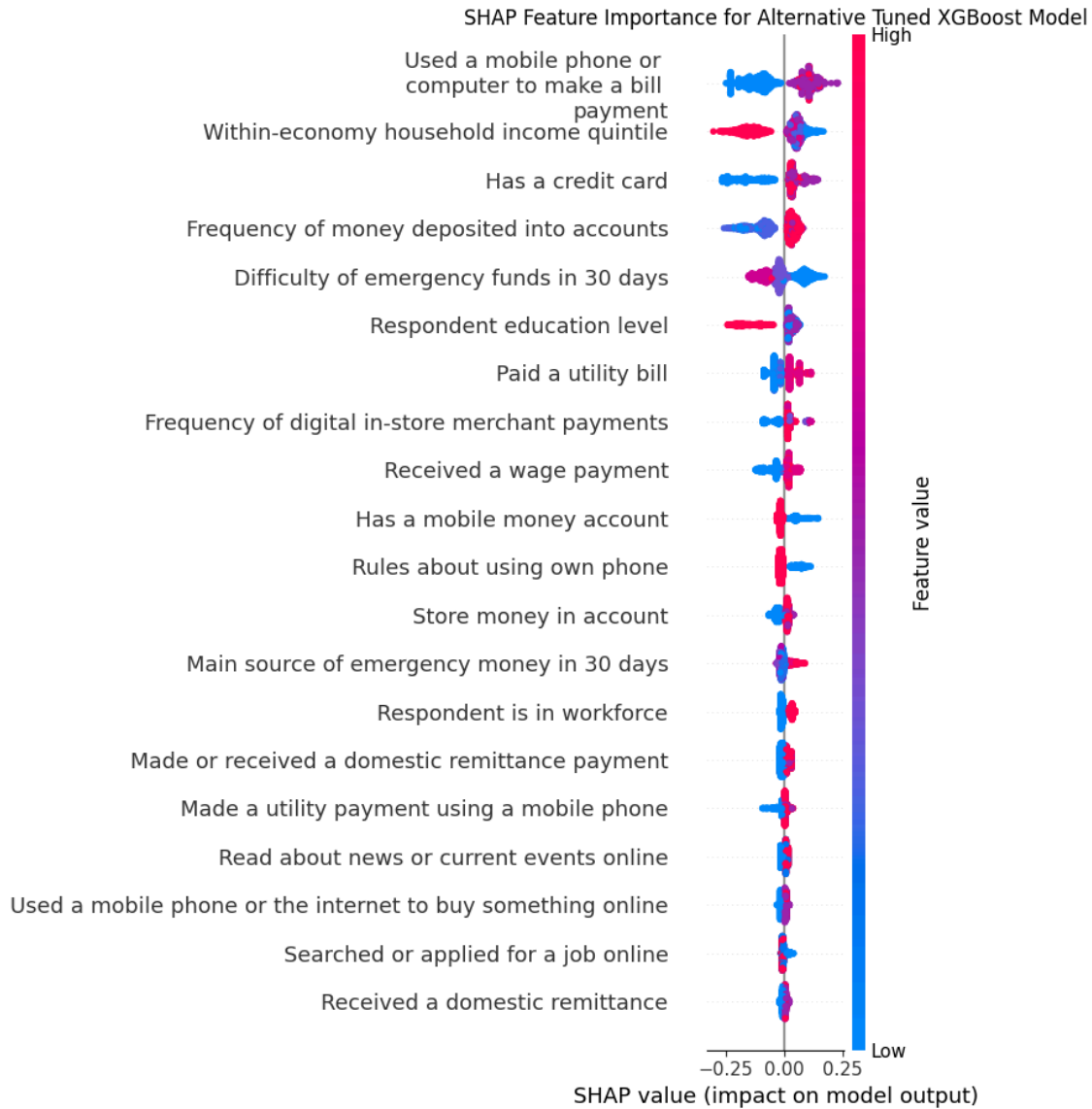
The XGBoost model was retrained with the same recall-driven grid, which settled on a 0.01 learning rate, a max depth of 3 and 200 estimators.

Table 6. Full model versus inclusion variant, decline-class metrics on the holdout

Model	Precision	Recall	F1	Accuracy
Tuned XGBoost, all features	0.417	0.726	0.530	0.661
Tuned XGBoost, non-traditional only	0.348	0.608	0.443	0.597

Decline recall fell from 0.726 to 0.608, but, still better than the 26% share of declines in the dataset.

This alternative model's **SHAP** profile shows the intended shift towards alternative predictors.

**Figure 11.** SHAP profile of alternative XGBoost model

Alternative digital footprint and behavioral indicators now dominate the list of predictors: paying bills with a mobile phone or computer became the top predictor, followed by income quintile, credit card ownership, deposit frequency and emergency-fund liquidity, with utility payments, digital merchant payments and mobile money account ownership also prominent.

This variant offers banks a real financial-inclusion option, but, at a measurable cost in performance.

VI. Summary, Conclusion & Next Steps

Philippine rural banks serve the bigger banks pass over, yet, tech constraints keep them from using alternative credit data. This project built a **proof-of-concept model to show what alternative data can do**. Such a model can benefit rural banks via a 10% approval uplift at about ₱74 million and a 2-point reduction in past-due loans at about ₱15 million

With no access to proprietary bank records, the **World Bank Global Index 2025** survey was used. 9,691 records with loan applicants were isolated from the entire 144,090-case dataset.

A four-step screen cut 199 variables to **78 predictors**: removing prohibited and irrelevant attributes, dropping sparse columns, grouping rare values and testing association with chi-square.

Three classifiers (**L1 Logistic Regression, CatBoost and XGBoost**) were assessed on their ability to catch declined applicants, each tuned to maximize decline recall. **XGBoost** won narrowly with decline recall of 0.726 and F1 of 0.530, a conservative profile that suits lenders protecting a small portfolio.

The audit exposed the model's main weakness – it prioritizes prior borrowing and income, which disadvantages unbanked applicants. A variant built only on non-traditional features kept the decline recall at 0.608 and shifted the model's attention to digital bill payments, deposits and mobile money. This gives banks a clear choice between maximum risk capture and broader inclusion.

6.1 Next Steps

To move from a global proof of concept to a governed local pilot, the following steps are recommended (Figure 12):

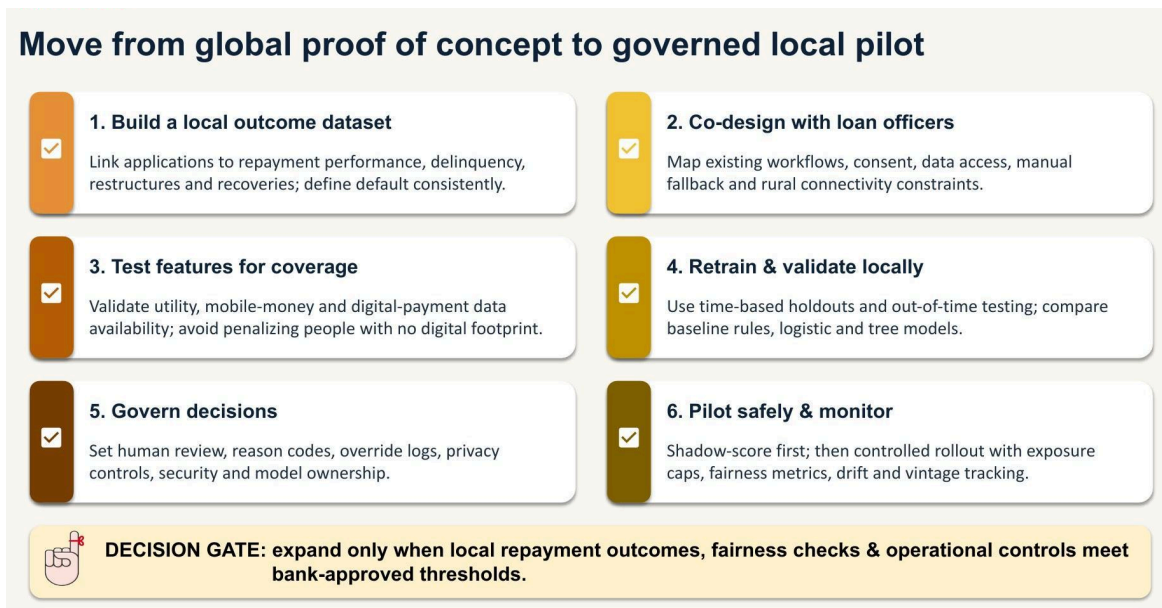


Figure 12. Move from global proof of concept to governed local pilot

One **decision gate** governs the whole path. The bank expands use of the model only when local repayment outcomes, fairness checks and operational controls meet thresholds the bank itself has approved.

Annex A

DELIVERABLE	SUBMISSION DETAILS
Problem statement & target metric	- Slide 1-3 of "JOSEPH_HANS_Pillar_5_Capstone-BusinessDeck.pdf" - Section 1 of this document
Dataset overview & Data dictionary	"ddi-documentation-english_microdata-7860.pdf" "JOSEPH_HANS_Pillar_5_Capstone-DataDictionary.xlsx"
EDA & Feature Engineering Report	- Section 3 of "JOSEPH_HANS_Pillar_5_Capstone.ipynb" "findex_microdata_2025_loans-ImputationDefaults.csv" "rare_value_replacement.csv"
Trained models, metrics and comparison	- Section 4 of "JOSEPH_HANS_Pillar_5_Capstone.ipynb" "JOSEPH_HANS_Pillar_5_Capstone-best_lr_model.pkl" "JOSEPH_HANS_Pillar_5_Capstone-best_cat_model.pkl" "JOSEPH_HANS_Pillar_5_Capstone-best_xgb_model_alt.pkl"
Bias & Fairness Analysis	- Section 5.iv of "JOSEPH_HANS_Pillar_5_Capstone.ipynb" - Section 5.3 of this document - Slide 7 of "JOSEPH_HANS_Pillar_5_Capstone-BusinessDeck.pdf"
Technical Deck Business Deck	"JOSEPH_HANS_Pillar_5_Capstone-TechnicalDeck.html" "JOSEPH_HANS_Pillar_5_Capstone-TechnicalDeck.pdf" "JOSEPH_HANS_Pillar_5_Capstone-BusinessDeck.pdf"
GitHub repo link Final Report Reproducible code	https://github.com/jopethans/JSH-AIML-Capstone - This document "JOSEPH_HANS_Pillar_5_Capstone.ipynb" "notebooks" & "data" folders in repo link
Running app, deployment guide & demo media	N/A
Document how GenAI was used	See Annex B of this document

Annex B: Use of Gen AI in this project

Annex B.1 Summarization and reformatting of material

LLMs were used to create a business deck draft, with final human review & refinement of the output.

STEP 1: Prompt to ChatGPT to convert the Jupyter Notebook into a presentation deck

```

please create a deck with 8 slides on a prototype alternative
credit scoring algorithm for rural banks. Output format is
powerpoint pptx. Use the uploaded "JOSEPH_HANS_Pillar_5
_Capstone.ipynb" as source, augmented with available online
material

SLIDE1: Technology Adoption Challenge for rural banks. Source
material data from "1. Problem Understanding & Framing" in the
ipynb

SLIDE2: Opportunity to use Machine Learning to jump-start
alternative credit scoring. Again, use "1. Problem
Understanding & Framing" in the ipynb

SLIDE3: Upside potential of alternative scoring; explain that
a rural bank can expand its loan portfolio by going beyond
traditional financial measures in evaluating loan
applications; create an illustrative loan asset increase by
getting average size of rural bank loan portfolio, assuming a
% increase from use of alternative scoring, calculating the
increase in loan portfolio from this % increase in approval
rates

```

Figure 13. Excerpt of ChatGPT prompt to create a business presentation deck

STEP 2: ChatGPT parses the Jupyter notebook and creates a basic set of slides

01 | TECHNOLOGY ADOPTION CHALLENGE

Rural banks face a technology adoption gap

Last-mile institutions. Lean technology capacity.
Rural banks serve remote and lower-income municipalities where mainstream banking may be limited—supporting farmers, fisherfolk, micro-entrepreneurs, local savings and government credit programs.

- Tight tech budgets**
Limited investment capacity for modern scoring systems.
- Connectivity gaps**
Poor or unreliable internet in remote areas.
- IT talent shortage**
Small teams with limited specialist capacity.
- Manual & legacy workflows**
Paper-heavy processes constrain data use and speed.

The consequence
Thin formal credit files can make otherwise viable rural borrowers difficult to assess.

Opportunity: use lightweight, customizable data tools to extend evaluation beyond traditional financial records—without replacing local credit judgment.

Source: Capstone notebook, §1. RBAP reference: rbap.org

01 / 08

Figure 14. Slide 1 of the presentation deck draft from ChatGPT

STEP 3: The ChatGPT deck loaded to Google Slides; Gemini prompted to generate graphics

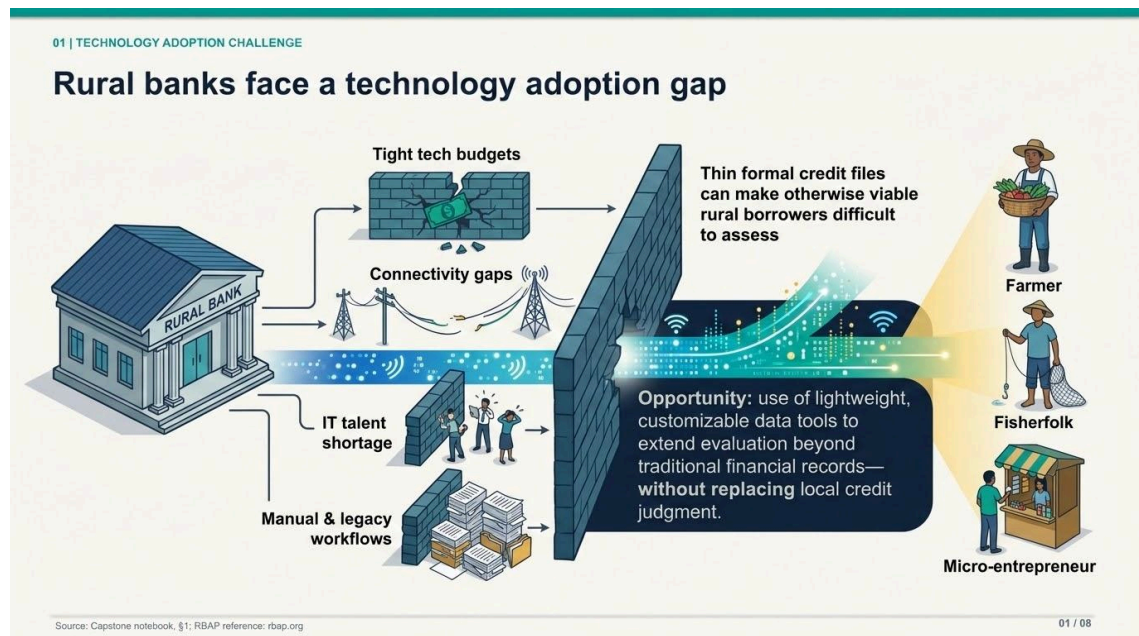


Figure 15. Slide 1 of the presentation deck with Gemini images

STEP 4: Manual edits done on the google slides to sharpen content and improve readability

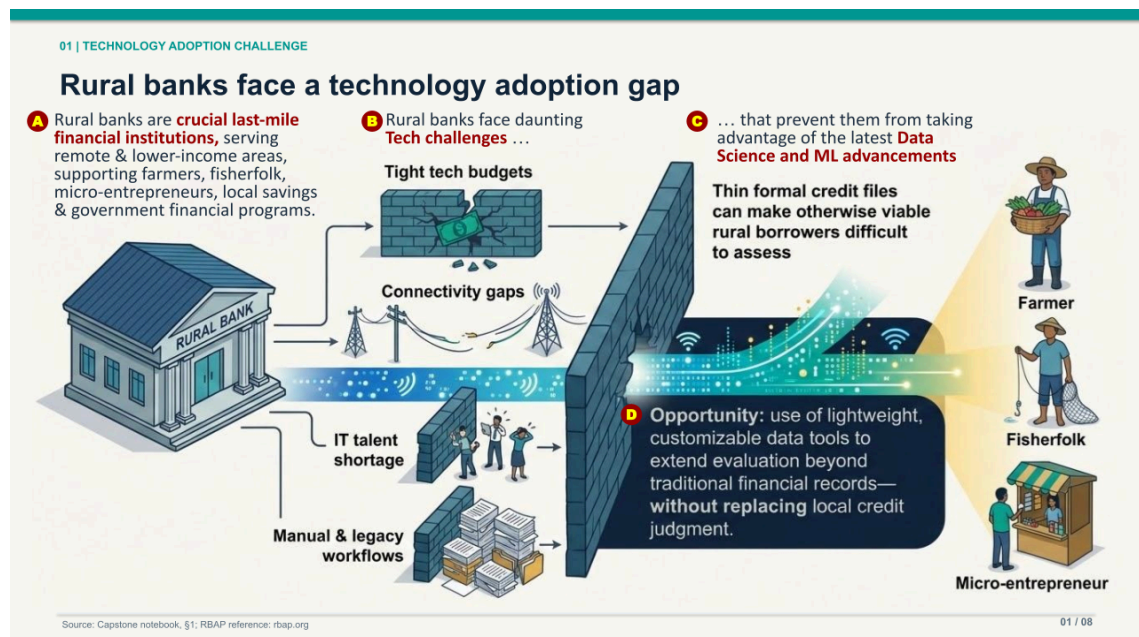


Figure 15. Final opening slide of business deck

Annex B.2 LLM Review of entire capstone project

LLMs was instrumental in reviewing the project and was able to detect issues that resulted in an overhaul of the project recommendation

STEP 1: Prompt to Claude to create final report from Jupyter notebook & the final deck from Annex B.1

Create a report summarizing the activities and results of a project to create a proof-of-concept alternative credit scoring model for Philippine Rural Banks. Output format is docx. Target length is 10 pages. Use the `Jupyter` notebook "JOSEPH_HANS_Pillar_5_Capstone.ipynb" as the main source. Supplement with graphics from the business deck "JOSEPH_HANS_Pillar_5_Capstone-BusinessDeck.pdf" Include the following chapters:

I. Problem statement: start with images of first slide from the business deck. Provide additional details from section 1 of the `Jupyter` notebook. Conclude this chapter with a discussion on business value, where you include slides 2, 3 & 4 of the business deck.

Figure 16. Prompt to Claude to create final report

STEP 2: Claude generates draft, AND reviews the entire project notebook noting critical discrepancies

The report is 10 pages of content plus a cover page with an "at a glance" results table. It follows your six chapters and slide placements, with slide 6 appearing at the top of both Chapter III and Chapter IV. I also added the notebook's SHAP plot in Chapter V next to the slide 7 audit, because it shows the feature ranking the text discusses.

While reading the notebook, I found three things you should check before submitting. I kept the report consistent with the notebook's own write-up, so none of these are fixed in the document.

1. **The feature count may be 84, not 81.** The funnel table says 81 features. The train/test split output, however, shows `X_train` with 84 columns, which is 87 minus `fin21`, `wpid_random` and `wgt`. That means the six columns that failed the chi-square test were never actually dropped before training. The inclusion-variant code supports this: it finds `fin17c` in `X`, and `fin17c` is one of the six that should have been removed. There's also a second issue: the arithmetic $199 - 21 - 91 - 6 = 81$ counts the ID, weight and target columns as features.
2. **The ₱16M figure on slide 4 uses a different base.** 2% of ₱737M is about ₱14.7M. The ₱16M only works out if you apply the 2 points to the post-uplift portfolio of roughly ₱811M. The report states that basis explicitly so the numbers reconcile. If you meant ₱737M, change the slide and the report to about ₱15M.
3. **The tuning optimized accuracy, not decline recall.** All three grid searches used `scoring='accuracy'`, even though the project's success metric is decline recall. The report describes this as it was done. Re-running the grid searches with `scoring='recall'` (for class 2) or `'f1'` would line the tuning up with the project's stated objective and with the "class-aware tuning" claim on slide 6.

Figure 17. Claude feedback on parsing through the project notebook

STEP 3: Addressing Claude's findings resulted in major overhaul of the project recommendations

MODEL EVALUATION: of the three tuned models, it is **Lasso Linear Regression** that has the best RECALL of the default/decline class. It's PRECISION is lowest, so there will be more "approve" cases that are misclassified as "decline". This makes the model more **conservative**, which is preferable, especially for small lenders who have limited resources, or, for banks that are just starting with alternative scoring and want to be careful in safeguarding their loan portfolio at the initial stages.

The **Tuned Lasso LR is recommended for alternative credit scoring** for Rural Banks.

Figure 18. Project conclusion prior Claude findings

MODEL EVALUATION: of the three tuned models, it is ****XGBoost**** that has the best RECALL of the default/decline class. It's PRECISION is lowest, so there will be more "approve" cases that are misclassified as "decline". This makes the model more **conservative**, which is preferable, especially for small lenders who have limited resources, or, for banks that are just starting with alternative scoring and want to be careful in safeguarding their loan portfolio at the initial stages.

The **Tuned XGBoost is recommended for alternative credit scoring** for Rural Banks.

Figure 19. Project final conclusion after addressing Claudes findings